Burnaby School-Area Traffic-Calming Coverage

Summary

This project looks at traffic-calming infrastructure around schools in Burnaby, British Columbia. I treated a municipal transportation-planning team as the client. The main goal was to find school areas with relatively low coverage that may need a closer look.

I used four City of Burnaby spatial datasets: 135 Places of Interest, 1,494 traffic-calming features, 269 traffic signals, and 3,229 intersections. The Places of Interest data contains 64 school-coded locations and 71 other public places. I used the other public places as a comparison group.

The median distance to the nearest traffic-calming feature was 97.8 m for schools and 170.2 m for comparison places. School coverage was 54.7% within 100 m, 87.5% within 250 m, and 98.4% within 500 m. Schools had higher coverage than comparison places at 100 m and 250 m. The 500 m difference was not statistically significant after correction for multiple tests.

I also used PCA and K-means to compare schools using six infrastructure features. The selected two-cluster solution had 41 schools with fewer nearby traffic-calming features and 23 schools with more. I think the distance, coverage, and clustering results work well together as a first screening of school areas.

1. Problem and client question

The main question is: **How consistently are recorded traffic-calming features distributed around Burnaby schools, and which school areas have relatively low coverage?**

I divided this into several smaller questions. How far is each school from the nearest traffic-calming feature, signal, and intersection? How many schools have a traffic-calming feature within 100 m, 250 m, or 500 m? Are schools different from other public places? Can schools be grouped into useful infrastructure profiles?

These questions use different course techniques on the same municipal-data problem. The project includes data collection and cleaning, spatial joins, non-parametric and proportion tests, visualization, PCA, and clustering.

2. Data collection and preparation

I downloaded all four layers from the City of Burnaby ArcGIS REST service on August 3, 2026. The downloader requested GeoJSON records and the metadata for each layer. It also handled pagination and checked that the downloaded count matched the source count. A manifest records the retrieval time, source URL, record count, spatial reference, and SHA-256 checksum for every file. None of the downloaded records had missing geometry.

Layer	Records	Role in the analysis
Places of Interest	135	64 school-coded locations and 71 comparison places
Traffic Calming Features	1,494	Nearest distances, threshold counts, and feature types
Traffic Signals	269	Nearest active-signal distance
Intersections	3,229	Nearest-intersection distance and map context

I selected schools where the City field PLACETYPE equals 500. The other 71 Places of Interest became the comparison group. I kept stable IDs and the fields needed for analysis. I decoded category values from the

layer metadata, changed blank strings to missing values, and treated one traffic-signal date before 1900 as a placeholder.

The metadata did not give names for 187 traffic-calming type records. I kept their original codes and labelled them as “Unlabelled code” instead of dropping them. I checked required columns, IDs, record counts, missing values, and geometry quality. Finally, I projected every layer to EPSG:26910 so distances could be calculated in metres.

3. Analysis methods

I used GeoPandas nearest spatial joins to find one nearest traffic-calming feature, non-removed traffic signal, and intersection for each school. I also calculated the distance from each school to every traffic-calming point. This gave the number of features within 100 m, 250 m, and 500 m. I counted a location as covered when at least one feature was inside the threshold. I repeated the nearest-feature and coverage calculations for the 71 comparison places.

The nearest-distance distributions were skewed, so I used a two-sided Mann–Whitney U test instead of a t-test. I also calculated a rank-biserial effect size. For coverage proportions, I used chi-square tests when every expected count was at least five. The 500 m table did not pass that check, so I used Fisher's test there. I used Holm correction because tested the three related distance thresholds.

For clustering, I joined six measurements for every school: nearest distances to traffic calming, signals, and intersections, and traffic-calming counts within 100 m, 250 m, and 500 m. I standardized the features because distances and counts have different units. I used PCA to show the data in two dimensions. I then compared K-means results for k = 2 through k = 5 using silhouette scores and cluster sizes. I used random_state=0 and n_init=10 so the result is reproducible.

4. Results

4.1 Distance and threshold coverage

The school median distance was 97.8 m. The comparison-place median was 170.2 m, so the median difference was 72.4 m. The Mann–Whitney result was U = 1459 and p = 0.00034. The rank-biserial effect size was -0.358. The negative sign means school distances tended to be smaller. Based on this data, the school and comparison-place distance distributions are different, with a moderate effect.

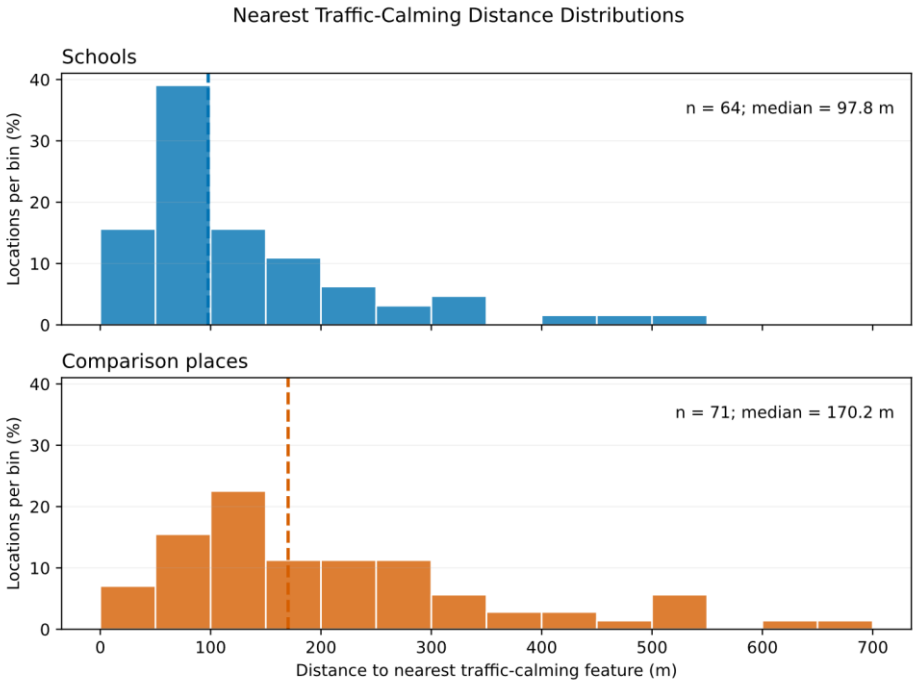


Figure 1. Nearest traffic-calming distance distributions. Both groups use the same 50 m bins and percentage scale. The dashed lines show the medians.

At 100 m, 35 of 64 schools were covered, or 54.7%. At 250 m, 56 schools were covered, or 87.5%. At 500 m, 63 schools were covered, or 98.4%.

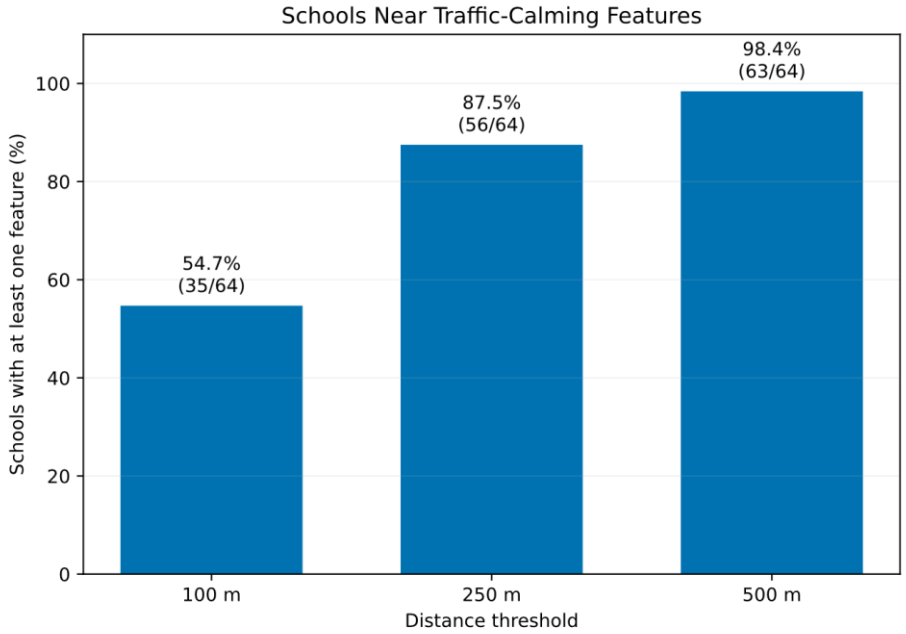


Figure 2. School traffic-calming coverage. A school is covered when at least one recorded traffic-calming feature is inside the distance threshold.

The school and comparison-place coverage rates were 54.7% and 22.5% at 100 m. The difference was 32.2 percentage points, with Holm p = 0.00073. At 250 m, the rates were 87.5% and 67.6%. The difference was 19.9 points, with Holm p = 0.022. I think these are clear differences in the current data.

At 500 m, the rates were 98.4% and 91.5%. Fisher's exact test gave Holm p = 0.119, so the two groups cannot be distinguished at this threshold. Coverage was already very high in both groups.

Speed bumps, speed humps, and barriers were the most widespread documented types around schools. Within 250 m, they appeared around 36, 30, and 25 schools, respectively.

4.2 School infrastructure profiles

The first PCA component explained 46.7% of the standardized feature variance, and the second explained 19.1%. Together they explained 65.8%. PC1 mostly contrasted higher traffic-calming counts with longer nearest-calming distance. The 250 m and 500 m counts were strongly correlated, with r = 0.849. I kept both because one describes closer coverage and the other describes the wider school area.

The silhouette scores for k = 2, 3, 4, 5 were 0.285, 0.268, 0.233, and 0.233. I selected k = 2 because it had the highest score and produced two groups of reasonable size.

Cluster 0 had 41 schools. Its median nearest traffic-calming distance was 139.6 m, and its median counts within 100 m, 250 m, and 500 m were 0, 3, and 11. Cluster 1 had 23 schools. Its median distance was 60.1 m, and its median counts were 3, 12, and 39. The median signal and intersection distances were almost the same in both clusters. This means the clusters are mainly separated by the distance to traffic calming and the number of nearby features.

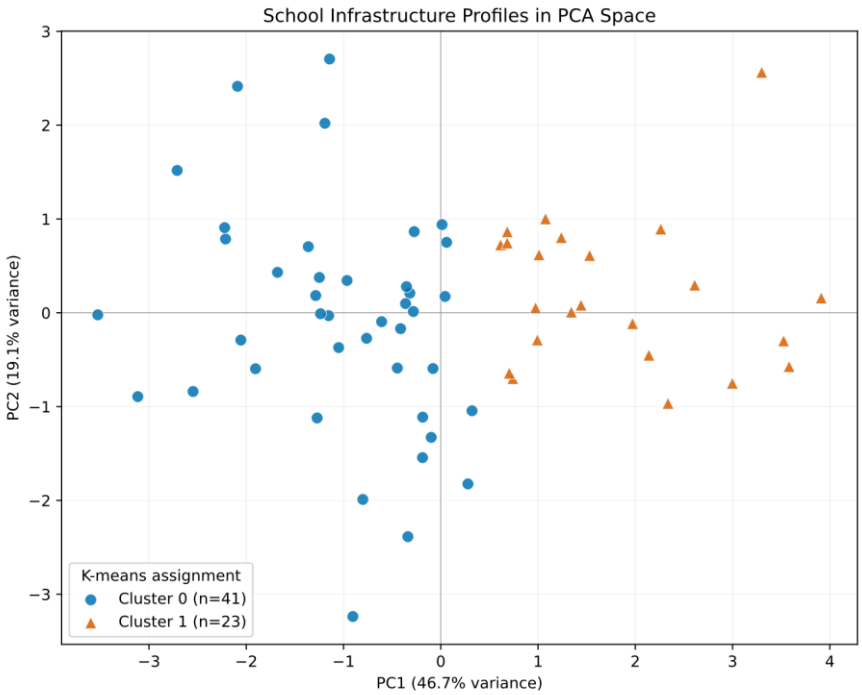


Figure 3. School infrastructure profiles in PCA space. The colours and marker shapes show the reproducible two-cluster assignments for 64 schools.

5. Conclusions and recommendations

More traffic-calming infrastructure is recorded near schools than near other public places, especially within 100 m and 250 m. However, the school results are not uniform. There were 29 schools with no feature within 100 m and eight with no feature within 250 m. The lower-coverage cluster also contained 41 of the 64 schools.

I think the most useful next step for the client is to review the exported school ranking, coverage counts, cluster assignments, and cluster map. Nearest distance and feature count should be considered together. A school can be close to one device but still have low coverage across the wider area.

6. Limitations and future work

The City data gives point locations but not the current condition of each device. It also does not include road alignment, traffic volume, vehicle speed, collisions, or student travel behaviour. Straight-line distance is not the same as a walking or driving route. Also, a larger feature count does not mean a school area is safer in reality.

The City's school category may include institutions outside a narrow K–12 definition. Some traffic-calming codes had no metadata label, and the municipal records may be missing, outdated, or changed after my download date.

The comparison group contains several kinds of public places. It was not matched to schools by neighbourhood or road environment. Nearby locations can also share the same traffic-calming features, so the observations are not fully independent. The statistical tests show associations, not that school status caused infrastructure placement. The clusters describe infrastructure patterns, not safety outcomes.

If I have more time, I will first calculate street-network distance. After that, I would add traffic volume, speed, collision, school-zone, and field-condition data. I think these additions would be more useful than adding another model to the current project.

Project Experience

For this project, I built an end-to-end Python geospatial pipeline using public data from the City of Burnaby. I collected and checked 5,127 municipal records, cleaned and projected the spatial layers, and used spatial joins, statistical tests, PCA, and K-means to study traffic-calming infrastructure around schools. The result I found most interesting was that schools had a median nearest-feature distance 72.4 m shorter than other public places, but coverage still varied a lot between individual schools. I enjoyed turning several raw municipal datasets into an analysis that a transportation-planning team could use as a starting point for further investigation.